SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: Cryptography and Network Security
COURSE CODE: CSA5117

COURSE OUTCOME COVERED

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 40%

QUESTIONS: 5 **Total Marks:100**

Annexure A

APPLICATION- BASED QUESTIONS

Code of Conduct:

I VEERLA NAGA RAVINDRA(192325067) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:Ravindra.v

Q. No	*Understanding of Concepts (25)*	*Experimental Design (30)*	*Use of Tools (20)*	*Analysis of Results (15)*	*Documentation (10)*	*Total Score (out of 100)*
1	*/ 5*	*/ 6*	*/ 4*	*/ 3*	*/ 2*	
2	*/ 5*	*/ 6*	*/ 4*	*/ 3*	*/ 2*	
3	*/ 5*	*/6*	*/ 4*	*/ 3*	*/ 2*	
4	*/ 5*	*/ 6*	*/ 4*	*/ 3*	*/ 2*	
5	*/ 5*	*/ 6*	*/ 4*	*/ 3*	*/ 2*	
OVERALL RUBRICS VALUE						
	/ 5	*/ 5*	*/ 4*	*/ 4*	*/ 2*	
	/ 25	*/ 30*	*/ 20*	*/ 15*	*/ 10*	*/ 100*

Annexure A

APPLICATION- BASED QUESTIONS

1.Scenario: Event Communication Security A student organization is preparing to share confidential event details with its core members using a simple classical encryption method. During testing, a message "MEET AT NOON" is transformed into an encoded version using a fixed shift pattern.
Reconstruct how the encrypted message might appear if a shift of 4 positions is applied.
Explain how an authorized member, aware of the encoding rule, would retrieve the original message.

2.Scenario: Intercepted Suspicious Message A cybersecurity intern intercepts the coded message: "GSRH RH Z HVXIVG" during network monitoring. The structure of the text suggests a simple substitution mechanism rather than complex encryption.
Deduce the most likely original message.
Justify your answer by analyzing repeating patterns, letter structures, or known cipher characteristics.

3.Scenario: Secure Messaging System A company deploys a keyword-based encryption technique for internal communication. An employee sends the word "HELLO" using the keyword "KEY" to ensure confidentiality.
Demonstrate how the encryption process transforms the message step-by-step.
Explain how the repeating keyword influences each character of the ciphertext.

4.Scenario: Suspicious Digital Asset Investigation During a digital forensic audit, investigators suspect that a company logo file may contain hidden confidential information embedded within it. The file appears normal upon casual inspection.
Analyze how investigators could detect the presence of hidden data within the image.
Recommend two techniques that could strengthen the security of such hidden communications against detection or extraction.

5.Scenario: Legacy Encryption in Corporate Communication A legacy system in a company still uses a transposition-based encryption method with three levels of arrangement. A received message appears as: "WECRLTEERDSOEEFEAOCAIVDEN".

a. Determine the most probable original message by reversing the encryption process.
b. Critically evaluate why such a method is vulnerable in modern communication systems.
c. Propose one practical enhancement to improve the confidentiality of messages in this scenario.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding g; limited or partial integration	Poor understanding g; unable to integrate concepts
Experiment al Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation on with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation

Q1. Event Communication Security (Caesar Cipher)

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Encryption

Plaintext: **MEET AT NOON**

Shift = **4**

Plain Letter	Encrypted Letter
M	Q
E	I
E	I
T	X
A	E
T	X
N	R
O	S
O	S
N	R

Ciphertext: QIIX EX RSSR

Decryption

The authorized member knows the key (shift = 4).

- Shift every encrypted letter **4 positions backward**.
- Q → M
- I → E
- X → T
- E → A
- R → N
- S → O

Recovered message:

MEET AT NOON

Conclusion: Caesar Cipher encrypts data by shifting letters. Decryption uses the reverse shift to recover the original message

Q2. Intercepted Suspicious Message

Ciphertext:

GSRH RH Z HVXIVG

This is encrypted using the **Atbash Cipher**, where each letter is replaced by its opposite in the alphabet.

Example:

- G → T
- S → H

- R → I
- H → S

Similarly,

- RH → IS
- Z → A
- HVXIVG → SECRET

Original Message

THIS IS A SECRET

Justification

- Repeated pattern "RH" becomes "IS".
- "HVXIVG" correctly converts to "SECRET".
- Atbash uses fixed alphabet reversal and does not require a key.

Conclusion: The intercepted message is **"THIS IS A SECRET."**

Q3. Secure Messaging System (Vigenère Cipher)

Plaintext:

HELLO

Keyword:

KEY

Repeat keyword:

```
Plaintext : H E L L O
Keyword   : K E Y K E
```

Using Vigenère encryption:

Plain	Key	Cipher
H	K	R
E	E	I
L	Y	J
L	K	V
O	E	S

Ciphertext

RIJVS

Explanation

- The keyword repeats continuously.
- Each keyword letter decides the shift amount.
- Different shifts make frequency analysis more difficult than Caesar Cipher.

Conclusion: The Vigenere Cipher provides stronger security by using multiple shifting values instead of one fixed shift.

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Q4. Suspicious Digital Asset Investigation (Steganography)

How investigators detect hidden data

1. Perform statistical analysis of pixel values.
2. Compare the image size with similar images.
3. Use steganalysis software (StegExpose, StegDetect).
4. Examine Least Significant Bits (LSB) for unusual changes.
5. Compare the image with the original version if available.

Two techniques to improve security

1. Encrypt the secret message before embedding

- Use AES encryption before hiding the data.
- Even if extracted, the information remains unreadable.

2. Randomized or Adaptive LSB Embedding

- Hide data in randomly selected pixels instead of sequential pixels.
- Makes detection by steganalysis much harder.

Conclusion: Combining encryption with secure steganography greatly improves confidentiality and resistance to detection.

Q5. Legacy Encryption in Corporate Communication (Rail Fence Cipher)

Ciphertext:

WECRLTEERDSOEEFEAOCAIVDEN

(a) Original Message

Using the **3-Rail Fence Cipher** decryption:

WE ARE DISCOVERED FLEE AT ONCE

(b) Why this method is vulnerable

1. No secret mathematical key.
2. Limited number of rail combinations.
3. Easily broken using brute-force attacks.
4. Letter frequencies remain unchanged.
5. Modern computers can decrypt it within seconds.

(c) Practical enhancement

Use **AES-256 encryption** (or another modern encryption algorithm) instead of a simple transposition cipher.

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Benefits:

- Strong confidentiality.
- Resistant to brute-force attacks.
- Widely used in banking, government, and secure communication.

Conclusion: Replacing the Rail Fence Cipher with AES-256 provides significantly stronger security for modern communication systems.